SPARROW Mini Bill of Materials

Hardware sourcing checklist for the SPARROW Mini Assembly and Setup Guide

Source basis: SPARROW Mini Setup Guide RC1.docx, V1.0

Purpose
This BOM sits beside the SPARROW Mini setup guide so builders can source every item needed for one complete unit and identify optional sensor, deployment, and legacy assembly choices.

Build assumptions

- The standard one-unit build uses a Raspberry Pi Zero 2 W or WH, microSD storage, a Voltaic 20 W solar panel, a Voltaic V75 battery with built-in solar controller, a self-powered Micro-USB OTG Ethernet/USB hub with a dedicated external 5 V input, and a weatherproof field enclosure.
- A RELAY / DigiMesh unit also requires one compatible XBee 868 MHz or 900 MHz radio, a UART-to-USB adapter, and a band-matched antenna. A DIRECT-only unit may omit the radio path when it has Wi-Fi or 4G internet access.
- Quantities are minimum recommended quantities for one SPARROW Mini. Items marked per sensor, as needed, or site dependent scale with the selected camera, microphone, enclosure, and mounting method.
- Cable jacket color is not a requirement. Select cables by connector type, electrical rating, data capability where needed, length, and environmental rating.

Core BOM for one standard SPARROW Mini unit

System	Component	Qty	Status	Use in SPARROW Mini	Sourcing notes
Compute and storage	Raspberry Pi Zero 2 W or WH	1	Required	Runs SPARROW Mini services and the edge AI workload; interfaces with radio and sensors.	Use the Pi Zero 2 W or WH model named in the guide.
Compute and storage	microSD card (32Gb recommended)	1	Required	Primary operating-system and local data storage.	Use a high endurance card compatible with Raspberry Pi Zero 2 W; choose capacity for the expected image/audio retention.
Compute and storage	Compatible Raspberry Pi Zero 2 W/WH power supply	1	Required for setup	Provides stable bench power during imaging, first boot, and troubleshooting.	Match the Pi Zero 2 W micro-USB power input and manufacturer electrical requirements. You can also use the Voltaic V75 recommended below.
Power system	Voltaic 20 W, 6 V solar panel	1	Required for field build	Charges the field battery for autonomous operation.	Use the manufacturer cable/plug intended for the selected Voltaic battery; do not alter terminals.
Power system	Voltaic V75 USB battery pack, 20,100 mAh	1	Required	Stores solar energy and powers the Pi; includes the solar charge controller described in the guide.	Use the V75 or a fully compatible pack with the required USB output and solar input.
Power system	USB-A to Micro-USB power cable	1	Required	Connects the battery USB output to the Pi Zero 2 W PWR IN port.	Must be power-capable, correctly terminated, and long enough for strain-free routing. Color is irrelevant.
Connectivity	Self-powered Micro-USB OTG Ethernet adapter/hub with USB-A ports and dedicated 5 V power input	1	Required	Connects the Pi USB data port, provides Ethernet, and powers multiple USB peripherals without drawing their full load through the Pi OTG port.	Procure a Pi Zero 2 W-compatible OTG hub with RJ45, enough USB-A data ports, a separate external 5V input, ideally select one with a documented current capacity, over-current protection, and upstream-port power isolation or a documented no-backfeed design.
Connectivity power	External 5 V hub power cable	1	Required	Connects the selected 5 V source to the hub's dedicated power-input connector.	Match the exact hub connector and polarity; use a short, adequately gauged power-only or manufacturer-approved cable. Do not use an improvised Y-cable.
Connectivity power	Regulated 5 V supply or independent battery USB output for hub	1 output	Required	Supplies the hub and attached XBee/sensor peripherals at their combined peak current.	The selected V75 output arrangement has enough independent capacity; otherwise use a separate regulated source. Keep the Pi PWR IN and hub power paths isolated. You can use the 2 USB ports in the V75 to independently power the Pi PWR IN and the Hub.
Connectivity	Ethernet cable and active network connection	1	Required when using wired Ethernet	Provides wired connectivity during setup or operation when that path is used.	Cable category and length may vary; use an outdoor-rated cable if exposed.

System	Component	Qty	Status	Use in SPARROW Mini	Sourcing notes
Radio	Digi XBee SX 868 MHz (EU) or XBee-PRO 900 MHz (US) RF module	1	Required for RELAY / DigiMesh	Provides long-range low-power DigiMesh or point-to-multipoint communication.	Select the legal frequency family for the deployment country; all mesh nodes must use compatible radios.
Radio	SparkFun Digi XBee Explorer, USB A or USB-C, WRL-11697 or WRL-22043, or verified equivalent	1	Required for RELAY / DigiMesh	Holds a through-hole XBee module and provides USB-C UART communication to the powered hub.	WRL-11697 are a verified current model with USB-A or or WRL-22043 with USB-C UART and an onboard regulator rated up to 2 A.
Radio cable	USB-A to USB-C data cable	1	Required with WRL-22043	Connects SparkFun Digi XBee Explorer USB-C WRL-22043 to the powered hub.	Use a short data-capable cable rated for the adapter and radio peak current.
Radio	Band-matched XBee antenna	1	Required for RELAY / DigiMesh	Provides the RF path for the XBee module.	Match the antenna to 868 MHz or 900/915 MHz, the radio connector, local rules, and outdoor use. Using High gain antennas will allow for longer range connectivity.
Radio	RF adapter/pigtail or RP-SMA hardware	As needed	Required if connectors differ	Mates the selected XBee module to the selected antenna without stressing the board.	Purchase only if the radio and antenna do not connect directly; verify connector gender and polarity.
Enclosure and mounting	IP65 or NEMA 4X weatherproof enclosure with backplate or sealed cable-entry cover box	1	Required for outdoor deployment	Protects the Pi, battery, radio, and internal cabling from rain, dust, and interference.	The guide permits a weatherproof extension-cord cover box or a junction box with perforated backplate.
Enclosure and mounting	Cable-entry seals or glands for enclosure penetrations	1 set	Required if not included	Seals the solar, antenna, Ethernet, and sensor cable entry points.	Size each seal/gland to the actual cable and enclosure opening; retain the enclosure rating.
Enclosure and mounting	Double-sided mounting tape	1 pack	Required	Secures the Pi to the USB adapter and the Pi/adapter assembly to the battery as shown.	Use a bond suitable for the device surfaces and field temperatures.
Enclosure and mounting	Internal mounting method: VHB tape, adhesive hook-and-loop, or heavy-duty zip ties	1 set	Required	Secures the complete battery, Pi, adapter, and XBee assembly to the enclosure backplate.	At least one listed method is required; select for serviceability, vibration, and the enclosure material.
Setup peripherals	Host PC with internet access	1	Required for setup	Runs Raspberry Pi Imager, downloads software, and starts device configuration.	Windows, macOS, or Linux can be used.
Setup peripherals	microSD card reader	1	Required for setup	Connects the microSD card to the host PC for imaging.	Built-in or external is acceptable if compatible with the card and host PC.
Setup peripherals	Available electrical outlets	At least 2	Required for setup space	Powers the host PC and Pi/power system in parallel during setup.	Additional outlets may be needed for tools or optional peripherals.

Optional legacy assembly options

System	Component	Qty	Status	Use in SPARROW Mini	Sourcing notes
Legacy power control	USB MOSFET power switcher	1	Optional legacy	Provides software-controlled switching of a USB-powered legacy accessory.	Use only for the legacy switched-accessory path. Match its Micro-USB input, USB-A output, logic level, continuous current and peak current to the accessory.
Legacy power control	Data-capable Micro-USB power-input cable for MOSFET switcher	1	Required with MOSFET	Supplies the switcher from the selected regulated 5 V source.	Match connector orientation and current rating; do not assume a charge-only cable is suitable where data/control is required.
Legacy power control	USB-A output cable or adapter for switched accessory	1	Required with MOSFET	Connects the switcher's output to the legacy accessory.	Choose the accessory-end connector only after confirming that accessory's input. Avoid stacked adapters where a direct cable is available.
Legacy power control	Female-female or matching Dupont jumper leads	1 set	Required with MOSFET	Connects Pi GPIO control and ground to the switcher.	Confirm the exact GPIO pinout, connector gender and logic level before connection; label both ends.
Legacy camera interface	Badgerd SDWire3 board	1	Optional legacy	Provides the legacy Badgerd SDWire3 camera/source path referenced by the guide.	Badgerd SDWire3. Procure only after confirming the target device, only allows SD or MicroSD as host interfacing. If the target device allows USB storage mode then, USB interfacing is recommended over SD/MicroSD.

System	Component	Qty	Status	Use in SPARROW Mini	Sourcing notes
Legacy camera interface	Full-size SD card ribbon extension	1 as needed	Conditional legacy	Relocates a full-size SD card slot to the SDWire assembly or accessible enclosure position.	Match full-size SD male/female ends, contact orientation, pin count and supported card mode. Keep the ribbon short, strain-relieved and away from sharp folds.
Legacy camera interface	microSD card ribbon extension	1 as needed	Conditional legacy	Relocates a microSD slot between the Pi/source device and the SDWire assembly.	Match microSD plug/socket orientation and device clearance. Verify that the extension supports the required signalling speed and does not obstruct the enclosure seal.
Legacy camera interface	microSD-to-full-size SD adapter	1 as needed	Conditional legacy	Adapts a microSD card or extension to a full-size SD interface.	Use only when the two selected components have different card form factors; verify write-protect-tab clearance and orientation.
Legacy storage	Dedicated SD or microSD storage card	1 per legacy storage path	Required with SDWire path	Stores images or data used by the legacy camera/source workflow.	Select the physical format, capacity, speed class and endurance rating required by the source device. Do not share the Pi boot card unless the legacy instructions explicitly require it.
Legacy camera interface	SDWire host data cable	1	Required with SDWire	Connects the USB-C port of the SDWire board to its host/control interface.	The available guide text does not establish the connector. Determine it from the exact SDWire revision and host before purchase; use a shielded data-capable cable where specified.

Optional sensor and deployment BOM

System	Component	Qty	Status	Use in SPARROW Mini	Sourcing notes
Mounting and deployment	Voltaic solar panel mounting plate	1	Optional	Mounts the solar panel and SPARROW Mini together for field deployment.	Can be secured with a strap or compatible screws/fasteners.
Mounting and deployment	Mounting strap, screw kit, anchors, or site fasteners	As needed	Site dependent	Secures the panel mount and enclosure to the selected support or fixed anchor.	Select hardware for the support material, wind load, corrosion exposure, and local requirements.
Mounting and deployment	Heavy-duty zip ties	Pack	Optional / site dependent	Provides alternate internal mounting and can secure the enclosure or panel to the mounting plate.	Use outdoor/UV-resistant ties for exposed deployments.
Radio upgrade	High-gain external XBee antenna	1	Optional upgrade	Extends practical radio range when site conditions and regulations allow.	Use a band-matched, weather-resistant antenna and compatible cable/connector hardware.
Camera sensing	Supported Reolink Wi-Fi camera	1 per camera unit	Optional	Captures images and sends alerts to the SPARROW Mini local network.	Reolink Altas or Argus Eco
Camera sensing	Camera battery, charger, or solar power accessory specified by the camera maker	1 per camera	Required with selected camera	Powers the selected Reolink camera in the field.	Match the selected camera model and follow its manufacturer instructions.
Camera setup	Mobile phone with Reolink app	1	Required for Reolink setup	Scans the camera QR code, joins it to the SPARROW hotspot, and configures email alerts.	Do not sign in to a Reolink cloud account during the offline setup described in the guide.
Audio sensing	AudioMoth DEV board	1 per audio unit	Optional	Provides the USB microphone/audio sensing path.	Flash the USB Microphone firmware before deployment.
Audio sensing	AudioMoth DEV case	1 per AudioMoth	Required with AudioMoth	Provides the field enclosure for the AudioMoth board.	The guide removes the battery cradle before installing the board and cable.
Audio sensing	Micro-USB cable for AudioMoth	1 per AudioMoth	Required with AudioMoth	Connects AudioMoth to a free USB-A port on the SPARROW Mini adapter/hub.	Use a data-capable cable; verify length and cable-gland fit. Color is irrelevant.
Audio sensing	PG9 cable gland	1 per AudioMoth case	Required with AudioMoth	Replaces the M16 placeholder cap and seals the Micro-USB cable entry.	Confirm the gland fits the case opening and selected cable.
Audio sensing	Velcro straps or zip ties	1 set per AudioMoth	Required for AudioMoth field mounting	Secures the AudioMoth case at its deployment location.	The case may include straps; otherwise source suitable outdoor ties/straps.

System	Component	Qty	Status	Use in SPARROW Mini	Sourcing notes
Direct connectivity	Wi-Fi or 4G internet connection and any required modem/router	1 connection	Required for DIRECT mode	Provides direct backhaul to the SPARROW server without DigiMesh forwarding.	The guide does not specify a modem model; select locally compatible service and hardware.

Tools and shop supplies

Tool or supply	Use
Anti-static mat or non-conductive work surface	Protects the Pi, XBee, microSD, and adapters during assembly.
Marker or pencil	Marks enclosure/backplate mounting points and cable-entry locations.
Basic hand tools	Small screwdrivers, pliers, scissors or flush cutters, and other tools needed by the selected enclosure and fasteners.
Electric drill and suitable bits	Needed only when enclosure or mounting holes must be created or enlarged.
Precision knife	Used only if an AudioMoth Micro-USB connector must be carefully trimmed to pass through the PG9 cable gland.
Protective eyewear and gloves	Recommended personal protective equipment for assembly and installation.
Fire safety item	Keep a fire blanket, container of sand, or suitable extinguisher within reach when working with batteries.
Approved fall-protection equipment	Required when installation, inspection, or servicing is performed at height.

Software, accounts, and setup dependencies

Dependency	Purpose
Raspberry Pi Imager	Writes the operating-system image to the microSD card.
Raspberry Pi OS Lite, 64-bit	Operating system selected in the guide for Raspberry Pi Zero 2 W.
SPARROW Mini setup script / sparrow-client repository	Installs required software, creates the local hotspot, configures the XBee family, and starts the service.
Temporary Wi-Fi credentials and secure SPARROW Mini hotspot password	Needed for the initial installation and local device network.
GitHub personal access token, if the repository is private	Needed only when private repository access is required during installation.
Internet access	Needed for Raspberry Pi Imager, package/repository downloads, and optional firmware/apps.
SPARROW server address, API key, and onboarding email	Required only when onboarding a DIRECT-mode unit to the SPARROW server.
AudioMoth Flash App and USB Microphone firmware 1.3.1	Required only for the AudioMoth option described in the guide.
Reolink mobile app	Required only when configuring a Reolink Wi-Fi camera.

Procurement and substitution notes

- Confirm local electrical, radio, environmental, and installation compliance before purchasing parts.
- Match the XBee radio, antenna frequency, antenna connector, and regulatory approval to the deployment country.
- Select cables by connector, electrical/current rating, data capability where required, environmental rating, and length. Cable color is only a visual reference in the guide.
- Use outdoor-rated enclosures, glands, fasteners, ties, and exposed cabling. Preserve the enclosure seal and provide strain relief.
- Confirm that the self-powered USB Ethernet hub provides enough data ports for the XBee adapter and every selected USB sensor. Add the hub and each peripheral's peak current, include at least 25% headroom, and verify the external 5 V source, cable and connector ratings. Use only a hub with documented upstream power isolation or no-backfeed behavior.
- Use named brands/models where the guide or verified manufacturer information establishes compatibility. Where no exact model is established, buy against the stated connector, electrical, protocol, environmental and mechanical requirements rather than assuming a visually similar part is compatible.
- For legacy SDWire and MOSFET assemblies, confirm the exact board revision and target-device connectors before ordering ribbon extensions, adapters, data cables and power leads. The available guide text does not establish a universal legacy wiring specification.
- Plan cable lengths and connector clearances before mounting parts. Leave service slack without creating pinch points.
- Follow the battery, solar panel, camera, and enclosure manufacturers' safety and warranty instructions. Do not alter battery or solar terminals.

Safety and compliance notice

Builder responsibility

SPARROW Mini is an open-source, end-user assembled system. Builders are responsible for selecting compliant components, following manufacturer instructions, using proper electrical and mounting practices, and ensuring the completed unit meets local electrical, radio, environmental, occupational-safety, and installation requirements.